

## Unit-2

### Task-02

Update the ubuntu container, install Apache and php, then push updated on to docker hub.

Compare the size of base image and updated image after commit.

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\VICTUS> docker tag ubuntu kavya4619m/ab3:v1.0
PS C:\Users\VICTUS> docker login
Authenticating with existing credentials... [Username: kavya4619m]

Info -> To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
PS C:\Users\VICTUS> docker run -it --name ab3 kavya4619m/ab3:v1.0
root@f6b870a05210:/# apt update
Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2386 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3788 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2142 kB]
```

```
root@f6b870a05210:/# apt install -y php
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  adduser apache2 apache2-bin apache2-data apache2-utils ca-certificates
  krb5-locales libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3
  libaprutil1-ldap libaprutil1t64 libargon2-1 libbrotli1 libbsd0 libcurl4t64
  libedit2 libexpat1 libgdbm-compat4t64 libgdbm6t64 libgssapi-krb5-2 libicu74
  libjansson4 libk5crypto3 libkeyutils1 libkrb5-3 libkrb5support0 libldap-common
  libldap2 liblua5.4-0 libnghttp2-14 libperl5.38t64 libpsl5t64 librtmp1 libsasnl2-2
  libsasnl2-modules libsasnl2-modules-db libsodium23 libsqlite3-0 libssh-4
  libssl3t64 libxml2 media-types netbase openssl perl perl-modules-5.38 php-common
  php8.3 php8.3-cli php8.3-common php8.3-opcache php8.3-readline psmisc
  publicsuffix ssl-cert tzdata ucf
Suggested packages:
  liblocale-gettext-perl cron quota ecryptfs-utils apache2-doc
  apache2-suexec-pristine | apache2-suexec-custom www-browser ufw php-pear
  gdbm-l10n krb5-doc krb5-user libsasnl2-modules-gssapi-mit
  | libsasnl2-modules-gssapi-heimdal libsasnl2-modules-ldap libsasnl2-modules-otp
```

```
info: Executing deferred 'a2enmod php8.3' for package libapache2-mod-php8.3
Enabling module php8.3.
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of start.
Setting up php8.3 (8.3.6-0ubuntu0.24.04.8) ...
Setting up php (2:8.3+93ubuntu2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for ca-certificates (20240203) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Processing triggers for php8.3-cli (8.3.6-0ubuntu0.24.04.8) ...
Processing triggers for libapache2-mod-php8.3 (8.3.6-0ubuntu0.24.04.8) ...
invoke-rc.d: could not determine current runlevel
invoke-rc.d: policy-rc.d denied execution of restart.
root@f6b870a05210:/# exit
exit
PS C:\Users\VICTUS> docker commit ab3 kavya4619m/ab3:v1.0
sha256:6b728a50c04842e9e6d30e6a5a338805a97a4c5ec6b6169a5c1da533b5a1b4

RAM 3.49 GB CPU 0.00% Disk 19.17 GB used (limit 1006.85 GB)
```

```
docker:dind                c77e5d7912f9          520MB          141MB
ghcr.io/kavya4619mgupta/sql_1473:latest 17777bc09bc9          1.07GB          232MB
harpreet81/ap1:v1.1         59855d3dceb3           13MB           3.86MB
httpd:latest                dd178595edd6          177MB           47.6MB
kavya4619m/ab2:v1.0         cd1dba651b30          119MB           31.7MB
kavya4619m/ab3:v1.0         6b728a50c048          417MB           117MB
kavya4619m/ap1:v1.1         25109184c71b          13.1MB          3.95MB
kavya4619m/ap2:v2.1         69eae0896851          33.6MB          10.1MB
mongo:6                     03cda579c8ca          1.06GB          273MB
mongo:latest                6f55a078e025          1.27GB          333MB
my-docker-app2:latest       23efa06493e5          200MB           48.8MB
mysql:5.7                   4bc6bc963e6d          700MB           149MB
mysql:latest                6b18d01fb632          1.27GB          283MB
nginx:latest                c881927c4077          240MB           65.7MB
node-mongo-compose-app:latest c46eef56ee38          1.63GB          406MB
python11:latest             632a0c2f0891          210MB           51.1MB
smart-study-planner-app:latest b9dbfe9c2434          1.66GB          422MB
smart-study-planner-client:latest 2c8934d2ba1f          1.6GB           402MB
smart-study-planner-server:latest 63e9b5e6e23d          1.6GB           402MB

PS C:\Users\VICTUS> docker login
Authenticating with existing credentials... [Username: kavya4619m]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
PS C:\Users\VICTUS> docker push kavya4619m/ab3:v1.0
The push refers to repository [docker.io/kavya4619m/ab3]
924ad2284388: Pushed
a3629ac5b9f4: Mounted from library/ubuntu
v1.0: digest: sha256:6b728a50c04842e9e9e6d30e6a5a338805a97a4c5ec6b6169a5c1da533b5a1b4 size: 751
PS C:\Users\VICTUS>
```

**Step1:** Run ubuntu container with given image name and tag

**Step2:** Authenticate

**Step3:** Get inside the container

**Step4:** Update and install the PHP and Apache image in the container

**Step5:** exit

**Step6:** commit

**Step7:** compare base image and updated image

**Step8:** Push the image to docker hub